Ben Taylor

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Work Experience

Software Engineer Aug 2025 - Present

Barclays

- Developed and deployed a production Python-Kafka data pipeline to process real-time stream data (**>100 key metrics**) from Barclays' virtual platforms, enabling continuous model monitoring and data quality assessment.
- Built **observability dashboards** for metric, data-quality, and drift monitoring used by **>200** colleagues.
- **Led weekly calls** to support cross-functional teams in utilising observability products for **technical analysis**.

Data Science Intern Jun 2025 - Sep 2025

Collaboraite

- Developed end-to-end an **AI computer vision tool** for open-set entity matching digital forensics in law enforcement, applying advanced deep learning and computer vision methods that upon evaluation resulted in an **86.4% identification accuracy** (**29.3% increase** compared to baseline methods).
- Created in-depth documentation of the full development lifecycle and a novel dataset (**over 500,000 frame level ground truths**) facilitating research in open-set image candidate retrieval and user identification.

Data Science Intern Nov 2024 - Feb 2025

Avanti West Coast

- Performed in-depth **exploratory data analysis** and data cleaning with **R** on **over 300k** railway records, identifying driver-related incidents and severe regional flooding as leading causes of railway service delays.
- Presented findings via **multiple data visualisations** to stakeholders as part of a cross-functional analysis, **informing recommendations** on operational risk trade-offs.

Software Engineer Intern Jun 2024 - Sep 2024

Barclays

- Designed, built, and tested **microservice REST APIs** using Spring Boot creating a **scalable real-time system** that **automated** manual entitlement distribution across the bank.
- Achieved **over 90% coverage** via unit and integration tests in a TDD environment ensuring system reliability.

Projects

Illumi: Agentic AI Banking Assistant *Python, HTML, OpenAI API, LLM*

- Built a recommendation and insight agentic AI financial management assistant leveraging LLMs and prompt engineering that provides natural language insights and analysis into user spending patterns.

Dynamic Face Blurring Agent *Python, Machine Learning, Computer Vision*

- Engineered a generative AI pipeline that uses machine learning, OpenCV, MediaPipe, and FFmpeg to automatically detect and anonymise faces in video streams and producing an encoded output.

Habit Analysis App *Rust, Tauri, SQL, TypeScript*

- Developed a data-driven analytical web application which securely stores, tracks and analyses user habit data.

Education

Lancaster University Distinction (GPA: 4.0 equivalent)

Master of Science: Data Science *2024 - 2025*

Loughborough University First Class Honours (GPA: 4.0 equivalent)

Bachelor of Science: Computer Science (Honours) *2020 - 2024*

Diploma in International Studies from Nanyang Technological University Singapore *2022 - 2023*

Technical Skills

Coding & Tools: Python (NumPy, PyTorch, TensorFlow), Java (Spring Boot), R, SQL, C++, Excel, Kafka, CI/CD.

AI & ML: MLOps, Ranking Algorithms, Transformer Architecture, Agentic Workflows, Data Pipelines, Docker.